When Tokens Expand: End-to-End Adversarial Evaluation of LLM Compression

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Abstract

Language-model compressors replace a hand-designed predictor with learned next-token probabilities, but their system behavior depends on more than model accuracy. Tokenization is a variable-rate interface between source bytes and model symbols; probability quantization, arithmetic coding, dictionary transmission, and framing add further costs. We present a principled methodology for adversarially evaluating this complete pipeline. Our attacks preserve exact token and byte round trips while optimizing either surprisal per decoded byte or realized serialized size. Layer-by-layer accounting distinguishes model failure from coder and container overhead, and matched conventional codecs test whether failures are specific to learned compression. A Qwen2.5-0.5B evaluation finds that three canonical adversarial sequences require 5.73 times the ideal bits per token of matched text8. Post-hoc occurring-token masks reduce adversarial cost by 35.2%, but it remains 3.81 times the masked text8 cost. These results motivate source-normalized robustness metrics and a guarded codec that falls back to a conventional compressor when the learned path expands its input.

1 A learned predictor is only one system component

The systems problem is not merely to replace a hand-crafted predictor with an LLM. The operational object is a pipeline from UTF-8 bytes through tokenization, model inference, frequency quantization, arithmetic coding, dictionary transfer, and container framing. We ask whether untrusted inputs can make that complete system expand data, which layer causes the expansion, and how the system should respond. Our contribution is an end-to-end methodology with explicit byte-level contracts, system-aware attacks, cost attribution, and a bounded fallback path.

In this pipeline, tokenizer symbols represent unequal numbers of bytes and may decode context-dependently; finite frequency tables distort probabilities; a restricted vocabulary must be transmitted; and headers dominate small inputs. Thus the least likely token need not maximize compressed bits per source byte, and perplexity alone cannot characterize end-to-end robustness. We call tokenization a *compression interface*, not a self-contained compressor: the vocabulary and token-ID code must still be shared or transmitted.

Contributions. We make three systems contributions: (i) source-normalized metrics that separate model NLL, coder payload, dictionary transmission, and serialized size; (ii) canonical greedy and beam attacks that optimize decoded-byte or realized-code objectives; and (iii) a matched evaluation with random, natural-text, and conventional-codec controls. [TODO: Add the supported multi-model conclusion.]

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Scope. We do not claim that tokenization alone is a self-contained compressor, that approximate beam search finds a global worst case, or that sequence NLL is a universal lower bound for an individual string.

2 Tokenization as a compression interface

Let s be UTF-8 text, $T(s) = x_{1:N}$ its tokens, $D(x)$ its decoding, $h_t = x_{1:t-1}$, and $p_\theta(v \mid h_t)$ the model distribution. If the codec retains dictionary $A \subseteq V$, it renormalizes that distribution to $q_\theta(v \mid h_t, A)$. The realized sequence NLL is

$$L_A(x) = -\sum_{t=2}^{N} \log_2 q_\theta(x_t \mid h_t, A). \tag{1}$$

This is sequence cross-entropy, not Shannon entropy of the predictive distribution.

2.1 Source-normalized cost

Define the source-normalized ideal rate

$$R_{\mathrm{NLL}}(x; A) = \frac{L_A(x)}{8|D(x)|_{\mathrm{UTF8}}}. \tag{2}$$

For a realized arithmetic payload C_{AC}, fixed or transmitted dictionary cost C_A, and framing/header cost C_H, report

$$R_{\mathrm{payload}} = \frac{C_{\mathrm{AC}}}{8|D(x)|_{\mathrm{UTF8}}}, \tag{3}$$

$$R_{\mathrm{serialized}} = \frac{C_{\mathrm{AC}} + C_A + C_H}{8|D(x)|_{\mathrm{UTF8}}}. \tag{4}$$

Values above one denote expansion. The gap between R_{NLL} and R_{payload} measures coder/quantization cost; the remaining gap to $R_{\mathrm{serialized}}$ measures metadata and framing.

Objective mismatch. For candidate v, let $\Delta_b(v; h) = |D(h \,\|\, v)|_{\mathrm{UTF8}} - |D(h)|_{\mathrm{UTF8}}$. Ranking by $-\log p(v \mid h)$ need not match ranking by $-\log p(v \mid h)/\Delta_b(v; h)$: a 12-bit-surprise token adding four bytes loses to a 6-bit-surprise token adding one byte under the latter objective.

2.2 Canonicality and losslessness

Tokenizers need not satisfy $T(D(x)) = x$ for arbitrary token-ID sequences. An attack sequence is valid only if every accepted prefix obeys

$$T(D(x_{1:t})) = x_{1:t}. \tag{5}$$

The implementation examines candidates in objective order and accepts the first canonical extension. By induction, the completed sequence is the canonical token representation of the bytes supplied to every compressor. Final validation must also decode the serialized arithmetic-coded artifact and compare both token IDs and source bytes exactly.

3 Threat model and attacks

Adversary. Use a white-box adversary with access to tokenizer, logits, and coding algorithm. The adversary chooses a length-N canonical token sequence after a fixed seed, subject to a candidate alphabet G. Evaluate three alphabets separately: full non-special vocabulary, printable ASCII bytes, and canonical one-byte ASCII. The last isolates predictive failure from token-length variation; the broader alphabets expose the tokenizer interface itself.

Attacks. We compare seeded uniform valid tokens; greedy minimum probability; greedy maximum surprisal per added byte; beam search over cumulative NLL/source bit; and beam search that clones arithmetic-coder state and ranks finalized size. Byte increments decode the complete prefix before and after extension. We test the full non-special vocabulary, printable ASCII, and canonical one-byte ASCII. Beam search is approximate unless its branch factor covers all candidates; changing beam ancestry also requires disabling the KV cache.

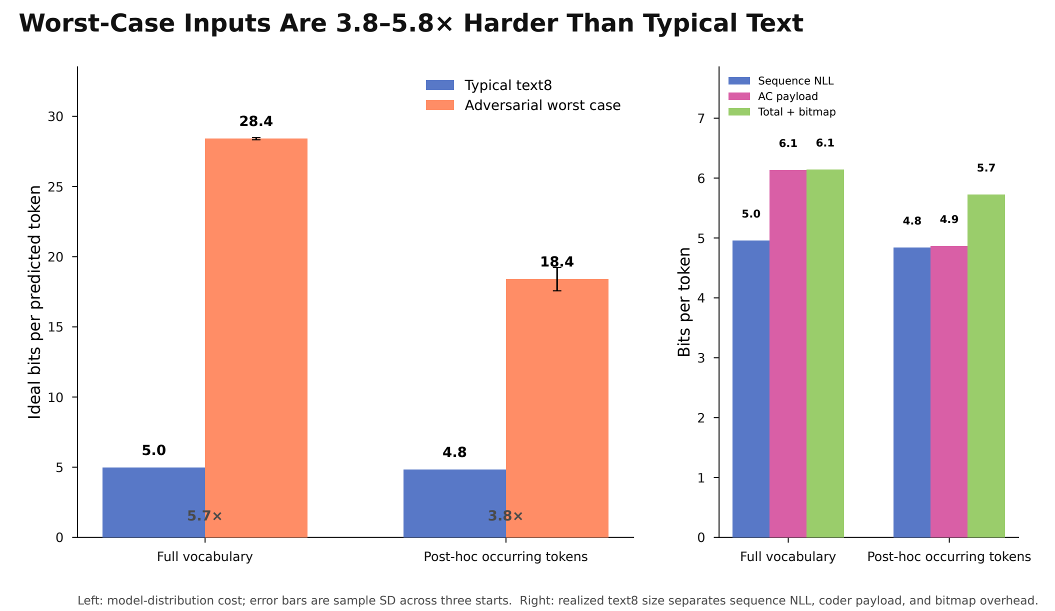

Figure 1: Notebook results for matched Qwen2.5-0.5B runs. Left: ideal sequence NLL per predicted token; adversarial bars show mean $\pm$ sample SD across three starts. Right: text8 NLL, realized arithmetic payload, and total including the dictionary bitmap. The adversarial result is idealized NLL, not a realized arithmetic payload.

Global mask. We construct each adversarial sequence once under the full distribution, then replay identical IDs and histories under its occurring-token mask. This isolates renormalization and is not a mask-adaptive attack.

4 Experimental evaluation

Setup. We evaluate Qwen2.5-0.5B with its 151,643-token vocabulary. The adversary creates three 1,000-token sequences from starting token IDs 785, 32, and 641 by repeatedly choosing the lowest-probability canonical extension. Every completed sequence satisfies $T(D(x)) = x$ and decodes to 6,988, 6,812, and 6,943 UTF-8 bytes. We then replay the identical token IDs and contexts using the set of tokens occurring in each completed sequence; the three post-hoc dictionaries contain 592, 606, and 587 tokens. Thus masking changes only probability normalization, not generation.

The natural-text control is the first 100,000 text8 tokens, processed by the same model with context length 1,000, retained context 100, KV caching, and arithmetic coding. It uses 16 independently seeded batches, so 99,984 tokens are predicted. We report mean adversarial surprisal with sample standard deviation across the three starts. Text8 is one deterministic matched run per dictionary policy and therefore has no error bar.

Worst case versus typical text. Figure 1 shows that full-vocabulary adversarial inputs require 28.409 ± 0.077 ideal bits per predicted token, compared with 4.954 for text8—a $5.734\times$ increase. The adversarial sequences use on average 595 distinct tokens. Restricting each fixed sequence to its occurring-token dictionary lowers its cost to 18.402 ± 0.833 bits/token, a 35.2% reduction, but it remains $3.806\times$ the matched text8 cost of 4.835 bits/token. Identical full-model scores under both policies verify that the token IDs and contexts did not change.

For text8, the full-vocabulary arithmetic payload is 6.136 bits per predicted token and the total is 6.138 bits per source token. The occurring-token policy reduces these to 4.860 and 5.724 bits, respectively, despite transmitting a 10,818-byte bitmap. Relative to raw UTF-8 size, the totals are 14.405% and 13.434%, corresponding to $6.942\times$ and $7.444\times$ compression. At 100,000 tokens, the occurring-token bitmap is 2.03% of raw size, so its coding gain exceeds its transmission cost.

Source-byte interpretation. The adversarial sequences have more UTF-8 bytes per token than text8. Relative to their own measured, round-tripping bytes, their mean model-induced floors are 51.346% for the full vocabulary and 33.284% after masking. These are floors for codes constrained to the stated model distributions; they exclude finite- coder and dictionary overhead and are not universal bounds for an individual string. This distinction is precisely why both token and source-byte views are needed.

Coder-aware validation. A preliminary 32-token one-byte-ASCII run provides a sanity check: token- and byte-greedy attacks coincide at 135.48% NLL and 143.36% payload, while an NLL beam reaches 136.70% and 144.14%. Because beam width is two and fixed overhead dominates, we use this only to validate objectives and accounting.

5 Systems implications and limitations

The systems lesson is that learned components need contracts at conventional boundaries. Source-normalized accounting identifies whether fragility comes from the tokenizer, predictor, quantizer, coder, dictionary, or container. For untrusted inputs, a guarded codec can store the smaller of learned and fallback encodings plus a selector bit, bounding size by the conventional codec plus one bit; its latency and throughput cost remains to be measured.

Our evidence is limited to one model, synthetic white-box inputs, three starts, finite context, approximate search, and one coder/container. It does not establish global worst cases or generalize to multilingual text or code. The attack could enable storage or compute amplification; practical mitigations include size caps, early-abort expansion detection, bounded inference, and fallback codecs.

6 Conclusion

Robust learned compression is an end-to-end systems problem. Canonical byte-aware attacks and explicit cost attribution show that the tested adversarial sequences are 3.81–5.73 times harder than matched text8, while guarded fallback offers a principled path from diagnosis to robust operation.

References

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